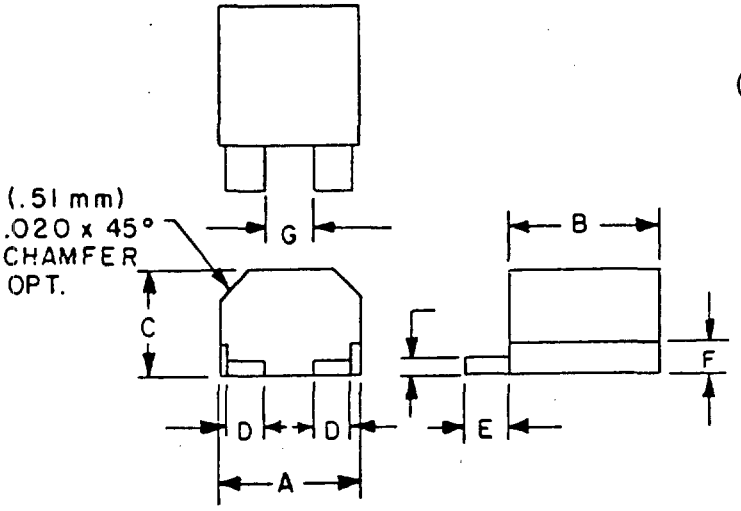
MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, FIXED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

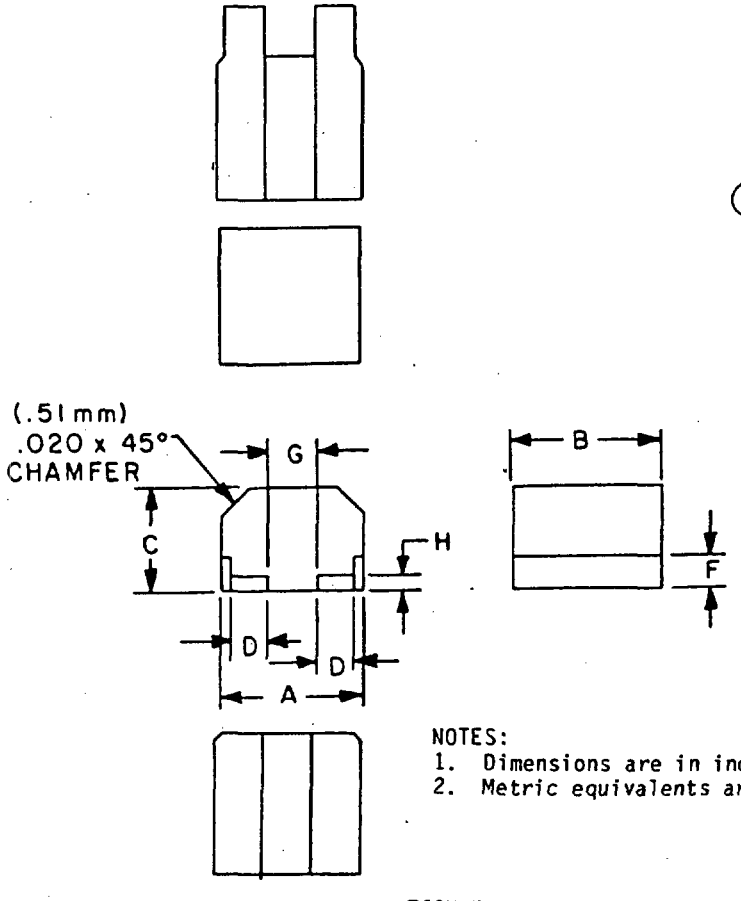
The requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-83446.



	Inches	mm
A	.110 Max	2.79 Max
B	.105 Max	2.67 Max
C	.065 Max	1.65 Max
D	.030 ±.005	0.76 ±0.13
E	.040 $^{+.015}_{-.010}$	1.02 $^{+0.38}_{-0.25}$
F	.018 ±.005	0.46 ±0.13
G	.038 ±.010	0.97 ±0.25
H	.005 ±.002	0.13 ±0.05

For dash numbers 01 thru 49

(With tabs)



	Inches	mm
A	.110 Max	2.79 Max
B	.105 Max	2.67 Max
C	.065 Max	1.65 Max
D	.030 ±.005	0.76 ±0.13
F	.018 ±.005	0.46 ±0.13
G	.038 ±.010	0.97 ±0.25
H	.005 ±.002	0.13 ±0.05

For dash numbers 50 thru 98

(Without tabs)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Fixed, chip coil.

Ⓒ denotes changes

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 0.5 gram maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

© Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 300 volts rms.

Barometric pressure: Method 105, test condtion C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number	Inductance ±10%	Q (min)	Q (typ)	Test frequency 1/	Self-resonant frequency (min)	DC resistance (max)	2/ Current (max)
	µH			MHz	MHz	Ohms	mA
01, 50	0.010	60	70	150	2000	0.025	750
02, 51	0.012	"	"	"	2000	0.025	"
03, 52	0.015	"	"	"	1800	0.040	"
04, 53	0.018	"	"	"	1500	"	"
05, 54	0.022	51	60	100	1400	"	"
06, 55	0.027	51	60	"	1200	"	"
07, 56	0.033	47	55	"	1100	0.050	640
08, 57	0.039	"	"	"	1000	0.07	600
09, 58	0.047	"	"	"	900	0.08	550
10, 59	0.056	"	"	"	850	0.09	520
11, 60	0.068	"	"	"	840	0.10	480
12, 61	0.082	"	"	"	750	0.11	470
13, 62	0.100	"	"	50	580	0.11	470
14, 63	0.120	"	"	"	240	0.11	470
15, 64	0.150	"	"	"	230	0.12	450
16, 65	0.180	51	60	"	"	0.14	430
17, 66	0.220	"	"	"	"	0.20	350
18, 67	0.270	"	"	"	"	0.25	310
19, 68	0.330	"	"	"	200	0.30	280
20, 69	0.390	47	55	"	190	0.45	240
21, 70	0.470	47	55	25	180	0.50	230
22, 71	0.560	45	53	"	170	"	"
23, 72	0.680	"	"	"	160	"	"
24, 73	0.820	"	"	"	150	"	"
25, 74	1.00	"	"	"	130	"	"
26, 75	1.20	36	43	7.90	120	0.60	200
27, 76	1.50	36	43	"	100	1.1	160
28, 77	1.80	38	45	"	90	1.1	160
29, 78	2.20	38	45	"	85	1.5	130

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number	Inductance ±10%	Q (min)	Q (typ)	Test frequency 1/	Self-resonant frequency (min)	DC resistance (max)	2/ Current (max)
	µH			MHz	MHz	Ohms	mA
30, 79	2.70	41	48	7.90	80	1.7	125
31, 80	3.30	42	50	"	75	1.8	120
32, 81	3.90	42	50	"	65	2.0	110
33, 82	4.70	41	48	"	55	2.3	100
34, 83	5.60	41	48	"	45	2.6	98
35, 84	6.80	36	43	"	40	2.8	94
36, 85	8.20	"	"	"	35	3.0	90
37, 86	10.0	"	"	"	33	3.3	87
38, 87	12.0	"	"	"	26	4.0	79
39, 88	15.0	32	38	2.50	24	4.2	79
40, 89	18.0	"	"	"	21	4.4	75
41, 90	22.0	"	"	"	19	7.5	57
42, 91	27.0	"	"	"	14	8.0	55
43, 92	33.0	30	35	"	12	13.0	45
44, 93	39.0	"	"	"	10	17.0	38
45, 94	47.0	"	"	"	9.0	19.0	36
46, 95	56.0	"	"	"	8.5	23.0	33
47, 96	68.0	"	"	"	8.2	25.0	32
48, 97	82.0	"	"	"	8.0	28.0	30
49, 98	100.0	"	"	"	7.0	31.0	28

1/ Test frequency range: 25 through 150 MHz:

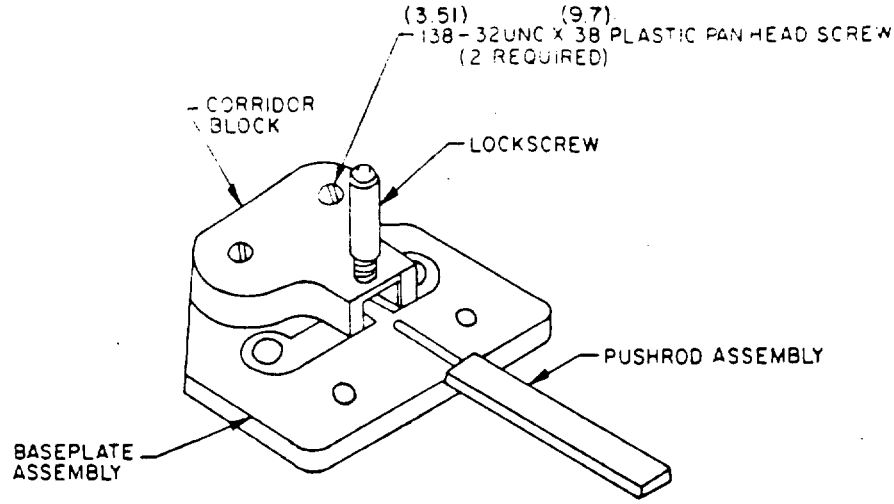
For electrical characteristics measurements use TF-250Q-1 test fixture, or equal (see figure 2). Fixture inductance (approximately 0.009 µH) should be subtracted from calculated inductance.

Test frequency range: 0.25 through 25 MHz:

For electrical characteristics measurements use TF-260Q-1 test fixture, or equal (see figure 3). Fixture inductance (approximately 0.028 µH) and residual Q-meter inductance (approximately 0.01 µH) should be subtracted from indicated inductance.

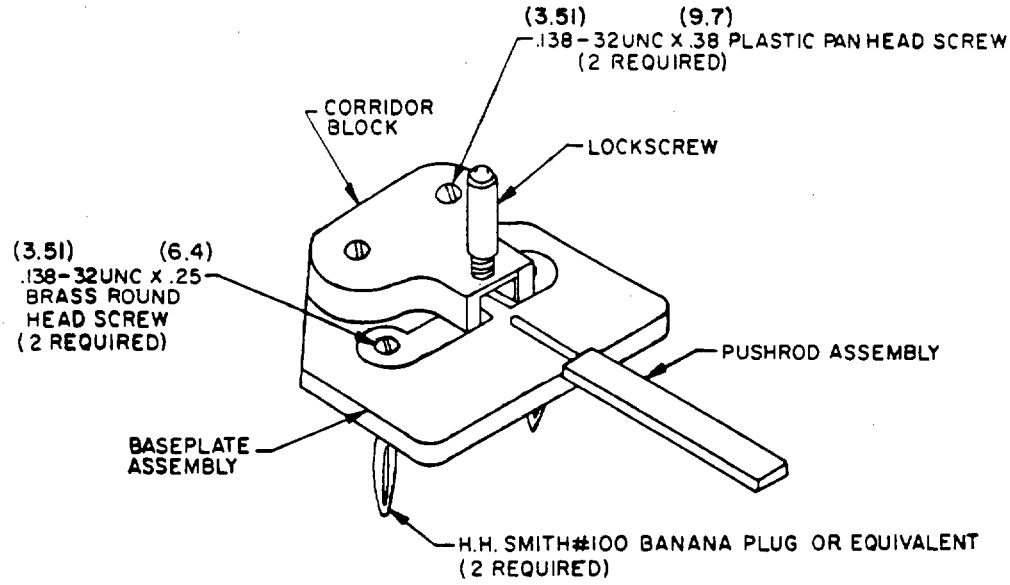
2/ Maximum current allowed not to exceed specified temperature rise.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 2. Typical chip coil test fixture (TF-250RX-1, or equal).



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 3. Typical chip coil test fixture (TF-260Q-1, or equal).

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from the initial measurements			
	Inductance	DC resistance	Self-resonant frequency	Q
Qualification inspection	Percent		Percent	Percent
Group II	±5	±(3% +0.001 ohm)	-8	-10
Group IV	±5	±(2% +0.001 ohm)	-10	-10
Group V	±2	---	---	-10
Quality conformance inspection				
Group C				
Subgroup II	±5	±(3% +0.001 ohm)	-8	-10
Subgroup IV	±5	±(3% +0.001 ohm)	-8	-10

Supersession data: A new part numbering system with codes for termination materials was incorporated with revision A, superseding MIL-I-83446/8, dated 1 March 1979.

M83446/08- (dash number from table I) C
supersedes M83446/8- (dash number from table I).

Examples:

M83446/08-21C supersedes M83446/8-21.
M83446/08-03C supersedes M83446/8-3.

Part number: The part number shall be in the following form.

M83446/08- 04 B

04: Sequentially assigned dash numbers (see table I)

B: Termination finish (see MIL-C-83446)

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS
Air Force - 19
DLA - ES

User activities:
Navy - OS
Air Force - 11

Preparing activity
Army - ER

Agent:
DLA - ES

(Project 5950-0699-3)

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